[INCH-POUND]
A-A-58076
02 August 1996
SUPERSEDING
MIL-H-4081B
21 AUG 1969

COMMERCIAL ITEM DESCRIPTION

CUTTER, COUNTERSINKS

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for cutters type I and type II, of MIL-H-4081B for all federal agencies.

1. **SCOPE.** This commercial Item Description covers countersinking cutters used on ferrous and non-ferrous materials.

2. **CLASSIFICATION.** The cutters shall conform to the following types and styles.

2.1	Type I	Countersink, plain taper shank
	Type II	Countersink, external threaded shank
2.2.	Style A	With integral constructed pilot.
	Style B	With attachable drill for pilot.

3. **SALIENT CHARACTERISTICS.**

3.1 Design. The cutters shall be made of heat treated high speed tool steel. The pilot diameter, taper shank or cutter centering angle, as applicable, and the cutting edges shall be concentric to 0.003 inch TIR. Configuration and dimensional requirements shall be as shown in figures 1 or 2 (as applicable) and as specified in Table I for combination drill and countersink using Style A pilots. Countersink cutters using Style B pilots may deviate from pilots sizes shown in Table I but shall have provisions for securing the pilots in place..

Beneficial comments, recommendations, additions deletions, clarifications, etc., and any data which may improve this document should be sent to: WR-ALC/TILCC 255 Second Street, Suite 122 Robins AFB, GA 31098-1640.

AMSC N/A FSC 5133

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3.2. Threads. Threads shall be 1/4 - 28 UNF - 2A. All threads shall be in the inch system and shall conform to the class 2 tolerance limits specified in the H28 Handbook.

3.3. Identification marking. The countersink shall be marked with the manufacturers name or identifying symbol, and the state or country of manufacture, unless otherwise specified. All markings shall be engraved, etched, molded, or indented directly on the items surface in such a manner that it remains clearly legible throughout the life of the item.

3.4. Workmanship. Details of workmanship shall be in accordance with the best commercial practice. Paints, coatings, platings, and finishes shall be smooth, dry adherent, continuous, and not stained or discolored. external surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections.

3.5. Cutting requirements. The cutter shall be capable of cutting repetitively when the cutter is mounted in a holder conforming to A-A-58075 using a portable, electric, or pneumatic powered hand tool. The holder and countersink shall be subjected to not less than ten repetitive countersink operation in stainless steel sheet, type 302, 8 gage minimum thickness. The countersink shall be at least 0.406 ± 0.005 in major diameter for all holes and the countersink surfaces shall not be rougher than 32 (AA). The included countersink angle and relation of an angle to the reference plane shall be accurate ± 1/2 degrees. Failure of the cutting requirement shall be cause for rejection of the representative lot.

4. **REGULATORY REQUIREMENTS.**

4.1. Recovered materials. The offerer/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation.

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they meet or exceed the requirements specified in this document.

5. **QUALITY ASSURANCE PROVISIONS.**

5.1. Contractor Conformance. The item delivered to the government shall conform to the contractually specified requirements. For item characteristics and requirements which are not contractually specified, the delivered item shall conform to the producers own drawings, specifications, standards, and quality assurance practices for the highest quality items provided by the producer to the commercial marketplace. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as provided for under the provisions of the contract.

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5.2. Market Acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

5.2.1. The company producing the item must have been producing a product meeting the requirements of this CID, for at least 2 years.

5.2.2. The company must have sold 1000 units meeting this CID, in the commercial market place over the past 2 years.

5.3. Performance Tests.

5.3.1. Visual and dimensional inspection. Each sample selected shall be visually and dimensionally inspected for conformance with this CID.

5.3.2. The countersink cutting test shall be performed with the cutter mounted in a holder conforming to A-A-58075, using a portable, electrical, or pneumatic powered hand tool. Holder and countersink shall be subjected to not less than ten repetitive countersink operations in stainless steel sheet, type 302, 8 gage minimum thickness. The countersink shall be at least 0.406± 0.005 in major diameter for all holes and the countersink surfaces shall not be rougher than 32(AA). The included countersink angle and the relation of angle to the reference plane shall be accurate to± 1/2 °. Failure of performance tests shall be cause for rejection of the representative lot. The number of cutters tested, and Lot size will be specified in the contract.

5.3.3. Additional requirements. Any additional requirements concerning responsibility for inspection, bid samples, or first article samples, shall be stated in the contract.

6. **PACKAGING.**

6.1. Packaging. Requirements for preservation, packaging, packing and marking of containers shall be as specified in the contract or order.

7. **NOTES.**

7.1. Addresses for referenced documents.

Federal Acquisition Regulation: Government Printing Office, Superintendent of Documents, Washington, DC 20402-9371.

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7.2. Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents:

a. Title, number and date of this Commercial Item Description.
b. When applicable, the appropriate NSN.
c. When applicable, the specific types, styles, and sizes (see Table 1).
d. The number of bid samples required.
e. Applicable first article sampling requirements.
f. Applicable lot sampling requirements.
g. Required preservation, packaging, and marking procedures.

7.3. National Stock Numbers (NSNs). The following NSNs are associated with this document. This list may not be indicative of all possible NSNs associated with this document.

SIZE	NSN 5133-00	TYPE	STYLE	CUTTER ANGLE
B	243-2298	I	A	100
C	243-2299	I	A	100
D	243-2300	I	A	100
M	293-2301	I	A	100
N	293-2303	I	A	100
O	595-8219	II	A	100
F	062-5290	II	A	100
F	062-5295	II	A	100
F	062-7216	II	A	100
A	177-7332	II	A	100
C	177-7334	II	A	100
D	177-7335	II	A	100

Custodians:
Air Force - 99
Army - GL
DLA - GS

Preparing Activity:
Air Force - 84

Reviewers:
Army - AV

Agent Activity:
Air Force - 99

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Project Number:
5133-0050

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TABLE I
Dimensional Requirements

Size	A - Pilot Exposed Length (Inches) ± 1/64	B - Body Diameter (Inches) ± 0.005	D - Pilot Diameter (Inches) + 0.000 – 0.001	C- Cutter angle ± 0° 30’
A	3/32	0.437	0.093	60°, 78°, 82°, 100°, 110°, 115°
B	1/8	0.437	0.125	
C	5/32	0.437	0.156	
D	3/16	0.437	0.187	
E	1/8	0.500	0.125	
F	5/16	0.500	0.250	
G	5/16	0.625	0.312	
H	3/32	0.375	0.093	
I	1/8	0.375	0.125	
J	5/32	0.375	0.156	
K	3/16	0.375	0.187	
L	1/4	0.500	0.156	
M	1/4	0.875	0.250	
N	1/4	0.875	0.312	
O	5/16	0.875	0.375	

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